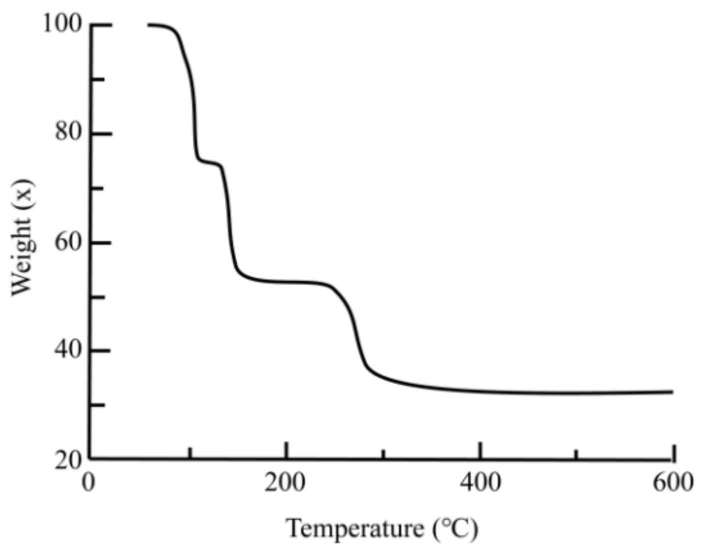
Question

A preparation method of a common metal complex is as follows: $1.8\ \mathrm{g}\ \mathbf{M}_2\mathrm{O}$, $50\ \mathrm{mL}\ \mathrm{CH_2Cl_2}$, $3\ \mathrm{mL}$ 3-hexyne (approximately $26.3\ \mathrm{mmol}$), $1.1\ \mathrm{g}$ oxalic acid, stirred for several hours, and the product was obtained by recrystallization with $\mathrm{CH_2Cl_2}$ at low temperature. It is known that the metal is a dinuclear complex, does not contain metal-metal bonds, and the metal content is 33.50%.

Thermogravimetric analysis of the complex was performed under inert conditions, and the results are shown in the figure below. Which of the following statements is incorrect?



This is a line graph. The abscissa is "Temperature($^\circ C$)", the scale division is 200, and the range is 0-600; the ordinate is "Weight(X)", the scale division is 40, and the range is 20-100. There is only one broken line in the figure, and there are the following inflection points (including the starting point): (70,100)(80,100)(100,75)(120,75)(130,55)(250,55)(280,30)(600,30)

A. All other options are incorrect

B. The product has the molecular formula $\mathbf{M}_2\mathrm{C}_{14}\mathrm{H}_{20}\mathrm{O}_4$

C. The product contains two five-membered rings.

D. The common salt solution of $\mathbf{M}$ is blue.

E. The final product of thermogravimetry is the element $\mathbf{M}$.

F. The first two thermogravimetric losses involve the same molecule.

Answer

Correct Answer: A

Detailed Explanation

According to the description in the question stem, "the metal is a dinuclear complex, does not contain metal-metal bonds, and the metal content is 33.50%", it is not difficult to guess that oxalate is a bridging ligand, coordinating with two $\mathbf{M}$. According to the preparation process, it can be seen that 3-hexyne participates in the reaction. It may be assumed that one $\mathbf{M}$ coordinates with one 3-hexyne. Therefore, the chemical formula of the product can be described as $\mathbf{M}_2(C_2O_4)(C_6H_{10})_2$. The molecular formula of the product is $\mathbf{M}_2C_{14}H_{20}O_4$, B is correct.

CHECKPOINT 1 PTS

The chemical formula of the product can be described as $\mathbf{M}_2(C_2O_4)(C_6H_{10})_2$

In the product, the oxalate radical acts as a tetradentate ligand, coordinating with two Cu respectively, and two 3-hexynes coordinate with Cu, forming two five-membered rings, C is correct.

CHECKPOINT 1 PTS

In the product, the oxalate radical acts as a tetradentate ligand, coordinating with two Cu respectively, and two 3-hexynes coordinate with Cu

According to the metal content, it can be judged that the metal is Cu, and the molecular weight of the product is $379.42\ \mathrm{g/mol}$. The common salt solution of $\mathbf{M}$ is blue, D is correct.

CHECKPOINT 1 PTS

The metal is Cu

The thermogravimetric image is discussed below. The first weight loss is about 25%, corresponding to about $95 \mathrm{~g/mol}$, corresponding to one 3-hexyne. Since it is reading the image, the actual weight loss is 21%. The second weight loss is about 25%, corresponding to about $95 \mathrm{~g/mol}$, corresponding to one 3-hexyne. The actual weight loss is 21%. The third weight loss is about 20%, corresponding to about $76 \mathrm{~g/mol}$, corresponding to 2 $\mathrm{CO_2}$, and the actual weight loss is 23%. It should be pointed out that the molecular weights of one 3-hexyne and two $\mathrm{CO_2}$ are relatively close. The judgment of the three weight losses takes into account the thermal stability of the coordination: π ligands are easier to leave than σ ligands. Therefore, E and F are both correct. Choose A.

CHECKPOINT 1 PTS

The first weight loss corresponds to 3-hexyne, the second weight loss corresponds to 3-hexyne, the third weight loss corresponds to 2 $\mathrm{CO_2}$, and the final residue is Cu